

2021

COMPUTER SCIENCE — HONOURS

Fifth Paper

8. (a) Discuss the three schema architecture for database systems.
 (b) Explain the process of translating the ER model into Relational model for one to many and many to many relationships.
 (c) What is a recursive relationship? Give a suitable example.
 (d) What is meant by data independence? Write about the two ways in which data independence can be achieved. 4+4+3+5
9. (a) Explain full outer join and left outer join with suitable examples.
 (b) Differentiate between sparse index and dense index.
 (c) Discuss the 'insertion anomalies', and 'deletion anomalies' with respect to normal forms with suitable examples.
 (d) What is lossless decomposition? 5+3+6+2
10. (a) For the following schema of the relational database:
 employee(ename, street, city)
 works(ename, company_name, salary)
 company(company_name, city)
 manages(ename, manager_name)
 Give an expression in sql for each of the queries given below :
 (i) Find all employees who live in the same city and on the same street as their managers.
 (ii) Find all employees in the database who do not work for SBI.
 (iii) Find all employees who earn more than every employee of PNB.
 (b) A database is being constructed to keep track of the teams and games of a sports league. A team has a number of players, not all of whom participate in each game. It is desired to keep track of the players participating in each game for each team, the positions they played in that game, and the result of the game. Design an ER schema diagram for this application, stating any assumptions you make. Choose any sport. (3×3)+7
11. (a) Suppose schema $R = (A, B, C, D, E)$ with $F = (A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A)$ is decomposed into (A, B, C) and (A, D, E) . Show that this decomposition is lossless-join decomposition.
 (b) Compute the closure for relation $r \{I, m, n, o, p\}$ with functional dependency set F as given below :
 $F = \{I \rightarrow mn; no \rightarrow p; m \rightarrow o; p \rightarrow I\}$
 Identify the candidate key for the relation (r) .
 (c) State some advantages and disadvantages of Normalization. 5+6+5